SUCCESSIVE DIFFERENTIATION & LEIBNITZ THEOREM

SUCCESSIVE DIFFERENTIATION

$$y = (ax+b)^n$$

$$y' = n(ax+b)^{n-1} \times a$$

$$y'' = a^2 n(n-1)(ax+b)^{n-2}$$

$$y''' = a^3 n(n-1)(n-2)(ax+b)^{n-3}$$

$$y'''' = a^4 n(n-1)(n-2)(n-3)(ax+b)^{n-4}$$

$$\therefore y^{b} = \frac{a^{b} n!}{(n-b)!}(ax+b)^{n-b}$$

$$
\begin{aligned}
&\underline{\underline{3.}} \quad y = \ln(ax+b) \\
&y' = \frac{1}{ax+b} \times a = \frac{a}{ax+b} = a(ax+b)^{-1} \\
&y'' = a \times -1\,(ax+b)^{-2} \times a = a^2 \times (-1)\overset{1!}{(ax+b)^{-2}} \\
&y''' = a^3 \times -1 \times -2 \times (ax-b)^{-3} = a^3 \times (-1)^2 \times \overbrace{1 \times 2}^{2!} \times (ax-b)^{-3} \\
&y'''' = a^4 \times -1 \times -2 \times -3 \times (ax-b)^{-4} = a^4 \times (-1)^3 \times \overbrace{1 \times 2 \times 3}^{3!} \times (ax-b)^{-4} \\
&y''''' = a^5 \times -1 \times -2 \times -3 \times -4 \times (ax-b)^{-5} = a^5 \times (-1)^4 \times \overbrace{1 \times 2 \times 3 \times 4}^{4!} \times (ax-b)^{-5} \\
&\therefore y^n = a^n \times (-1)^{n-1} \times (n-1)! \times (ax-b)^{-n}
\end{aligned}
$$

4. $y = \dfrac{1}{x+a}$

$$y = (x+a)^{-1}$$

$$y' = -1\,(x+a)^{-2} \quad = \quad (-1) \times 1 \times (x+a)^{-2}$$

$$y'' = -1 \times -2\,(x+a)^{-3} \quad = \quad (-1)^2 \times 1 \times 2 \times (x+a)^{-3}$$

$$y''' = -1 \times -2 \times -3 \times (x+a)^{-4} = (-1)^3 \times 1 \times 2 \times 3 \times (x+a)^{-4}$$

$$y^n = (-1)^n \times n! \times (x+a)^{-(n+1)}$$

$y = e^{ax}\sin bx$ show: $y_2 - 2ay_1 + (a^2+b^2)y = 0$

$y_1 = e^{ax}(\cos bx \times b) + \sin bx\,(ae^{ax})$

$= be^{ax}(\cos bx) + ae^{ax}\sin bx$

$y_2 = be^{ax}(-\sin bx \times b) + (\cos bx \times abe^{ax}) + ae^{ax}(\cos bx \times b) + \sin bx \times a^2e^{ax}$

$= -b^2e^{ax}\sin bx + abe^{ax}\cos bx + abe^{ax}\cos bx + a^2e^{ax}\sin bx$

Given $y = e^{ax}\sin bx$

$\therefore y_1 = be^{ax}\cos bx + ay \rightarrow \therefore be^{ax}\cos bx = y_1 - ay$

$y_2 = -b^2(y) + 2abe^{ax}\cos bx + a^2(y)$

$y_2 = -b^2(y) + 2a(y_1 - ay) \cancel{+a^2y}$

$y_2 = a^2y - b^2y + 2a(y_1 - ay)$

$\Rightarrow y_2 - a^2y + b^2y - 2a(y_1 - ay) = 0$

$\Rightarrow y_2 - a^2y + b^2y - 2ay_1 + 2a^2y = 0$

$\Rightarrow y_2 + a^2y + b^2y - 2ay_1 = 0$

$\Rightarrow y_2 - 2ay_1 + y(a^2+b^2) = 0$

$$y = \cos x$$

$$\frac{d}{dx}(y) = \frac{d}{dx}(\cos x)$$

$$= -\sin x$$

$$= \cos\left(\frac{\pi}{2}+x\right) \quad \cdots\cdots (1)$$

Derivative (1) w.r.t. x we get

$$\frac{d^2}{dx^2}(y) = \frac{d}{dx}\cos\left(\frac{\pi}{2}+x\right)$$

$$= -\sin\left(\frac{\pi}{2}+x\right)$$

$$= \cos\left[\frac{\pi}{2}+\left(\frac{\pi}{2}+x\right)\right] \quad \cdots\cdots (2)$$

Derivative (2) w.r.t x we get

$$\frac{d^3}{dx^3}(y) = \frac{d}{dx}\cos\left(2\cdot\frac{\pi}{2}+x\right)$$

$$= -\sin\left(2\cdot\frac{\pi}{2}+x\right)$$

$$= +\cos\left(\frac{\pi}{2}+2\frac{\pi}{2}+x\right)$$

$$= \cos\left(3\cdot\frac{\pi}{2}+x\right) \quad \cdots\cdots (3)$$

Proceding this way we can obtain

$$\frac{d^n}{dx^n}(y) = \cos\left(n\frac{\pi}{2}+x\right)$$

① find the nth derivative of '$\sin x$'

$y = \sin x$

Differentiating w.r.t 'x' we get.

$\boxed{Y_1 = \frac{dy}{dx}}$

$y_1 = \cos x$

Note $\quad Y_1 = \sin\left(\frac{\pi}{2} + x\right)$

Now $\quad Y_2 = \cos\left(\frac{\pi}{2} + x\right)$

$= \sin\left(\frac{\pi}{2} + \frac{\pi}{2} + x\right)$

OR $\quad Y_2 = \sin\left(2\frac{\pi}{2} + x\right)$

& $\quad Y_3 = \cos\left(2\frac{\pi}{2} + x\right)$

$= \sin\left(\frac{\pi}{2} + 2\frac{\pi}{2} + x\right)$

$Y_3 = \sin\left(3\frac{\pi}{2} + x\right)$

& $\quad Y_4 = \cos\left(3\frac{\pi}{2} + x\right)$

$Y_4 = \sin\left(4\frac{\pi}{2} + x\right)$

So we can say $\quad Y_5 = \sin\left(5\frac{\pi}{2} + x\right)$

$$\boxed{Y_n = \sin\left(n\frac{\pi}{2} + x\right)}$$

(2)

Q1. find nth Derivative of 'Sin (ax+b)'

Sol^n Given $Y = \sin(ax+b)$

Diff^n w.r.t 'x' we get-

Step1 $Y_1 = \cos(ax+b) \cdot a$

OR $Y_1 = \sin\left(1 \cdot \frac{\pi}{2} + (ax+b)\right) \cdot a$

diff^n again we get-

$Y_2 = \cos\left(\frac{\pi}{2} + (ax+b)\right) \cdot a^2$

Step 2 OR $Y_2 = \sin\left(\frac{\pi}{2} + \frac{\pi}{2} + (ax+b)\right) \cdot a^2$

$= a^2 \sin\left(2\frac{\pi}{2} + ax+b\right)$

diff^n again w.r.t 'x' we get-

$Y_3 = a^3 \cos\left(\frac{2\pi}{2} + ax+b\right)$

OR $Y_3 = a^3 \sin\left(\frac{3\pi}{2} + ax+b\right)$

from Step 1 & 2

So we can write nth Derivative as-

$$\boxed{Y_n = a^n \cdot \sin\left(n\frac{\pi}{2} + ax+b\right)}$$

Ans

LEIBNITZ THEOREM

$$\frac{d^n}{dx^n}uv = \frac{d^n u}{dx^n}v + n\frac{d^{n-1}u}{dx^{n-1}}\frac{dv}{dx} + \frac{n(n-1)}{2!}\frac{d^{n-2}u}{dx^{n-2}}\frac{d^2 v}{dx^2} + \dots$$
$$+n\frac{du}{dx}\frac{d^{n-1}v}{dx^{n-1}} + u\frac{d^n v}{dx^n}$$

$(uv)_n$ (differentiate n times)

$$(uv)_n = {}^nC_0 u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \ldots \ldots {}^nC_n u v_n$$

$$\left(\underbrace{x^3}_{u} \; \underbrace{\sin 4x}_{v}\right)_3$$

$$u = x^3$$

$$u' = u_1 = \frac{du}{dx} = 3x^2$$

$$u'' = u_2 = \frac{d^2u}{dx^2} = 6x$$

$$u''' = u_3 = \frac{d^3u}{dx^3} = 6$$

$$v = \sin 4x$$

$$v' = v_1 = \frac{dv}{dx} = 4\cos 4x$$

$$v'' = v_2 = \frac{d^2v}{dx^2} = -16 \sin 4x$$

$$v''' = v_3 = \frac{d^3v}{dx^3} = -64 \cos 4x$$

$$\therefore (uv)_3 = {}^3C_0 u_3 v + {}^3C_1 u_2 v_1 + {}^3C_2 u_1 v_2 + {}^3C_3 u v_3$$

$$= 1 \times 6 \times \sin 4x + 3 \times 6x \times 4\cos 4x + 3 \times 3x^2 \times -16 \sin 4x + 1 \times x^3 \times (-64 \cos 4x)$$

$$= \ldots$$

Leibniz Theorem

$$(x^2 e^{2x})_n = 2^n e^{2x} \cdot x^2 + {}^nC_1\, 2^{n-1} e^{2x} \cdot 2x + {}^nC_2\, 2^{n-2} \cdot e^{2x} \cdot 2$$

$$= 2^n e^{2x} x^2 + n\, 2^{n-1}\, 2x e^{2x} + \frac{n(n-1)}{2!}\, 2 \cdot 2^{n-2} e^{2x}$$

$$= 2^n x^2 e^{2x} + n\, 2^n x e^{2x} + n(n-1)\, 2^{n-2} e^{2x}$$

$$\boxed{{}^nC_r = \frac{n!}{r!\,(n-r)!}}$$

$${}^3C_2 = \frac{3!}{2!\,(3-2)!} = \frac{3!}{2!\,1!} = \frac{3\times2\times1}{2\times1\times1} = 3$$

$$y = x^2 e^{2x}$$

$$y' = 2x e^{2x} + 2x^2 e^{2x} =$$

$$y'' = 2e^{2x} + 4x e^{2x} + 4x e^{2x} + 4x^2 e^{2x}$$

$$= 2e^{2x} + 8x e^{2x} + 4x^2 e^{2x}$$

$$= 2(2-1)2^0 e^{2x} + 2 \cdot 2^2 x e^{2x} + 2^2 x^2 e^{2x}$$

$$y''' = 4e^{2x} + 8e^{2x} + 16x e^{2x} + 8x e^{2x} + 8x^2 e^{2x}$$

$$= 12e^{2x} + 24x e^{2x} + 8x^2 e^{2x}$$

$$= 3.(3-1)2^1 e^{2x} + 3.2^3 x e^{2x} + 2^3 x^2 e^{2x}$$

$$y^{iv} = 24 e^{2x} + 24 e^{2x} + 48x e^{2x} + 16x e^{2x} + 16x^2 e^{2x}$$

$$= 48e^{2x} + 64x e^{2x} + 16x^2 e^{2x}$$

$$= 4(4-1)2^2 e^{2x} + 4.2^4 x e^{2x} + 2^4 x^2 e^{2x}$$

$$\therefore y_n = n(n-1)2^n e^{2x} + n 2^n x e^{2x} + 2^n x^2 e^{2x}$$

$$= [n(n-1)2^n + n 2^n x + 2^n x^2] e^{2x}$$

$$\boxed{[x^3 \sin x]_n}$$

$$\frac{d^3}{dx^3}\left(\underset{u}{x^3}\ \underset{v}{\sin x}\right)$$

$u_0 = x^3$

$u_1 = 3x^2$

$u_2 = 6x$

$u_3 = 6$

$v_0 = \sin x$

$v_1 = \cos x$

$v_2 = -\sin x$

$v_3 = -\cos x$

${}^nC_1 = n$

${}^nC_2 = \frac{n(n-1)}{2!}$

${}^nC_3 = \frac{n(n-1)(n-2)}{3!}$

$$(uv)_3 = {}^3C_0\, u_3 v_0 + {}^3C_1\, u_2 v_1 + {}^3C_2\, u_1 v_2 + {}^3C_3\, u_0 v_3$$

$$= 6x \sin x + 3 \cdot 6x \cos x + 3 \cdot \cancel{6x}\, 3x^2 (-\sin x) + x^3(-\cos x)$$

$$= \cancel{6 \sin x + 9x^2 \cos x - 9x^2}$$

$$= 6 \sin x + 18x \cos x - 9x^2 \sin x - x^3 \cos x$$

If $Y = (ax+b).\sin x$ Then, Find $Y_n = ?$

Sol: Let $v = (ax+b)$ & $u = \sin x$

$\frac{dv}{dx} = v_1 = a$ & $u_n = \sin(x + n\frac{\pi}{2})$

$v_2 = 0$ $\quad u_{n-1} = \sin(x + (n-1)\cdot\frac{\pi}{2})$

$\vdots \quad u_{n-2} = \sin(x + (n-2)\frac{\pi}{2})$

$v_n = 0 \quad \vdots$

By Leibnitz theorem –

$$\frac{d^n}{dx^n}(uv) = u_n v + {}^nc_1\, u_{n-1} v_1 + {}^nc_2 \cdot u_{n-2} v_2 + \dots + {}^nc_n \cdot u \cdot v_n$$

$$\frac{d^n}{dx^n}(uv) = (ax+b)\cdot\sin(x + n\frac{\pi}{2}) + {}^nc_1 \cdot \sin(x + (n-1)\frac{\pi}{2})\cdot a + {}^nc_2 \cdot \sin(x+(n-2)\frac{\pi}{2})\cdot 0$$

$$= (ax+b)\cdot \sin(x + n\frac{\pi}{2}) + \frac{n!}{1!(n-1)!}\sin(x + (n-1)\frac{\pi}{2})\cdot a + 0$$

$$= (ax+b)\sin(x + n\frac{\pi}{2}) + na\cdot\sin(x + (n-1)\frac{\pi}{2})$$

BK

If $y=e^{m\cos^{-1}x}$ then show that

$$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+m^2)y_n=0$$

If $y=e^{m\cos^{-1}x}$ then show that

$$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+m^2)y_n=0$$

Ans. $y=e^{m\cos^{-1}x}$ ——— ①

$$y_1=\frac{dy}{dx}=\frac{d}{dx}e^{m\cos^{-1}x}$$

$$\Rightarrow y_1=e^{m\cos^{-1}x}\frac{d}{dx}(m\cos^{-1}x)$$

$$\Rightarrow y_1=e^{m\cos^{-1}x}.m.\frac{-1}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2}\,y_1=-me^{m\cos^{-1}x}$$

$$\Rightarrow \sqrt{1-x^2}\,y_1=-my \quad \text{(equation ①)}$$

Squaring both sides, $(1-x^2)y_1^2=m^2y^2$

Differentiating both sides,

$$(1-x^2)\frac{d}{dx}(y_1^2)+(y_1)^2\frac{d}{dx}(1-x^2)=m^2\frac{d}{dx}y^2$$

$$\Rightarrow (1-x^2)\,2y_1\frac{dy_1}{dx}+y_1^2(-2x)=m^2\,2y\frac{dy}{dx}$$

$$\Rightarrow (1-x^2)\,2y_1y_2+y_1^2(-2x)=m^2\,2y\,y_1$$

$$\Rightarrow 2y_1[(1-x^2)y_2-xy_1]=2y_1m^2y$$

$$\Rightarrow (1-x^2)y_2-xy_1=m^2y$$

$$\Rightarrow (1-x^2)y_2-xy_1-m^2y=0 \quad \text{——— ②}$$

Now we have to differentiate n times with respect to x, using Leibnitz theorem

$$(1-x^2)y_2 - xy_1 - m^2y = 0 \qquad \Big| \quad D = \frac{d}{dx},\ D^n = \frac{d^n}{dx^n}$$

$$\underbrace{D^n (1-x^2)y_2}_{1} - \underbrace{D^n(xy_1)}_{2} - \underbrace{D^n(m^2y)}_{3} = 0$$

① $D^n(1-x^2)y_2 = {}^nC_0 D^n y_2 (1-x^2) + {}^nC_1 D^{n-1} y_2 D(1-x^2)$

$\qquad + {}^nC_2 D^{n-2} y_2 D^2(1-x^2)$ $\qquad \boxed{2+n-1 = n+1} \quad \boxed{2+n-2 = n}$

$\qquad = 1.(1-x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2\times 1}(-2).y_n$

$\qquad = (1-x^2)y_{n+2} - 2nx\,y_{n+1} - n(n-1)y_n$

② $D^n(xy_1) = {}^nC_0 D^n y_1 x + {}^nC_1 D^{n-1} y_1 Dx$ $\qquad \boxed{1+n-1 = n}$

$\qquad = 1.\,y_{n+1}.x + n.y_n.1$ $\quad (1+n)$

$\qquad = xy_{n+1} + ny_n$

③ $D^n(m^2y) = m^2 D^n y = m^2 y_n$

$$[(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n] - [xy_{n+1} + ny_n] - m^2y_n = 0$$

$\Rightarrow (1-x^2)y_{n+2} - 2nxy_{n+1} - xy_{n+1} - n^2y_n + \cancel{ny_n} - \cancel{ny_n} - m^2y_n = 0$

$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$ (proved)

If $\log_e y = a\sin^{-1}x$ then, show that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

$$\log_e y = a\sin^{-1}x$$

$$\Rightarrow y = e^{a\sin^{-1}x}$$

$$\frac{d}{dx} y = \frac{d}{dx} e^{a\sin^{-1}x}$$

$$\Rightarrow y_1 = \frac{a}{\sqrt{1+x^2}} e^{a\sin^{-1}x}$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = a e^{a\sin^{-1}x}$$

$$\Rightarrow (1-x^2) y_1^2 = a^2 (e^{a\sin^{-1}x})^2$$

$$\Rightarrow (1-x^2) y_1^2 = a^2 y^2$$

$$\frac{d}{dx}[(1-x^2)y_1^2] = \frac{d}{dx} a^2 y^2$$

$$\Rightarrow (1-x^2)\, 2y_1y_2 - 2x y_1^2 = a^2 2yy_1$$

$$\Rightarrow (1-x^2) y_2 - xy_1 = a^2 y$$

Differentiating n times each terms

$$\frac{d}{dx_n}(1-x^2)y_2 - \frac{d}{dx_n} xy_1 = a^2 \frac{d}{dx_n} y$$

$$[(1-x^2)y_2]_n - [xy_1]_n = a^2 y_n$$

Applying Leibnitz's theorem in the above equation

$$[(1-x^2)y_2]_n - xy_1 = a^2 y_n$$

$$\Rightarrow \left[{}^nC_0\, y_{n+2}(1-x^2) + {}^nC_1\, y_{n+1}(-2x) + {}^nC_2\, y_n(-2)\right]$$

$$-\left[{}^nC_0\, x\, y_{n+1} + {}^nC_1\, y_n(1)\right] = a^2 y_n$$

$$\Rightarrow (1-x^2)y_{n+2} - 2nx\, y_{n+1} + \frac{n(n-1)}{\cancel{2!}} \times (-2)\, y_n$$

$$- x y_{n+1} - n y_n - a^2 y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - xy_{n+1}(2n+1) - [n^2 - \cancel{n} + \cancel{n} + a^2]\, y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - \cancel{x}(2n+1)\, xy_{n+1} - (n^2+a^2)\, y_n = 0$$

$$\boxed{y = \tan^{-1} x}$$

$$y_1 = \frac{1}{1+x^2}$$

$$(1+x^2) y_1 = 1$$

$$\frac{d}{dx}\left[(1+x^2) y_1\right] = \frac{d}{dx}(1)$$

$$\Rightarrow (1+x^2) y_2 + 2x y_1 = 0$$

Differentiating the above eq n times

$$\frac{d}{dx_n}\left[(1+x^2) y_2\right] + \frac{d}{dx_n}(2x y_1) = 0$$

$$\Big[\underbrace{(1+x^2)}_{v}\ \underbrace{y_2}_{u}\Big]_n + \big[2x\,y_1\big]_n = 0 \qquad n = \frac{d}{dx_n}$$

Applying Liebnitz Theorem

$$(uv)_n = {}^nC_0\, u_n v_0 + {}^nC_1\, u_{n-1} v_1 + {}^nC_2\, u_{n-2} v_2 + \cdots$$

$$\Rightarrow \Big[{}^nC_0\, y_{n+2}(1+x^2) + {}^nC_1\, y_{n+1}\, 2x + {}^nC_2\, y_n\, 2\Big] + \Big[{}^nC_0\, y_{n+1}\, 2x + {}^nC_1\, y_n\, 2\Big] = 0$$

$$\Rightarrow (1+x^2)\,y_{n+2} + 2nx\,y_{n+1} + \frac{n(n-1)}{\cancel{2!}}\cdot\cancel{2}\,y_n + 1\,y_{n+1}\,2x + 2n\,y_n = 0$$

$$\Rightarrow (1+x^2)\,y_{n+2} + 2x\,y_{n+1}(n+1) + n y_n[n-1+2] = 0$$

$$\Rightarrow (1+x^2)\,y_{n+2} + 2x(n+1)\,y_{n+1} + n(n+1)\,y_n = 0$$

(shown)

(3.) $y\sqrt{1-x^2} = \sin^{-1}x$

$\Rightarrow y^2(1-x^2) = (\sin^{-1}x)^2$

Differentiating both sides

$$(1-x^2)\,2yy_1 + y^2(-2x) = 2(\sin^{-1}x)\frac{1}{(1-x^2)^{1/2}}$$

$$\Rightarrow (1-x^2)\,2yy_1 + y^2(-2x) = \frac{2y\cancel{(1-x^2)^{1/2}}}{\cancel{(1-x^2)^{1/2}}}$$

$$\Rightarrow (1-x^2)y_1 - xy = 1$$

Differentiating n times

$$[(1-x^2)y_1]_n - [xy]_n = 0$$